

LifeVision

For an infant, one glance away can change everything.



Attention needs another layer

#1

Drowning is the leading cause of death for U.S. children ages 1–4

61%

of reported pool and spa fatalities under age 5 involved a gap in adult supervision

2×

as often, bystanders first noticed the victim compared with lifeguards in a 140-case series

These facts support layered protection. Lifeguards and responsible adults still matter.

Sources: CDC; CPSC 2024 report covering 2019–2021 fatalities; Pelletier & Gilchrist, Injury Prevention (2011)

One observable, one timer

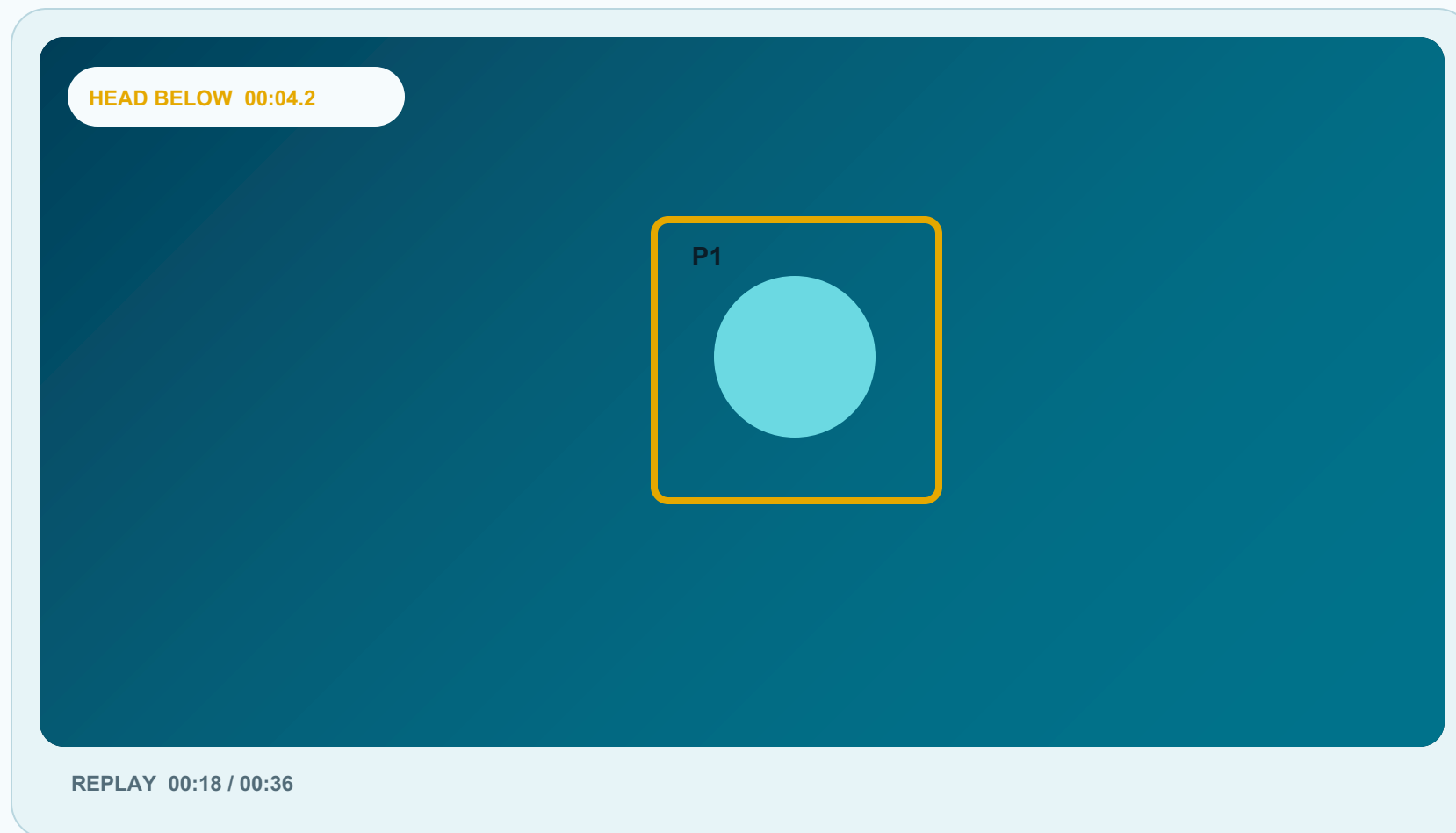
Life Vision measures head state. It does not diagnose drowning.



Yellow and red thresholds are user-configurable. Demo thresholds are shortened and labeled.

The interface makes the state change visible

Prototype interface shown as a scripted replay



DEMO MODE • shortened thresholds

ALERT STATE

YELLOW

Reason

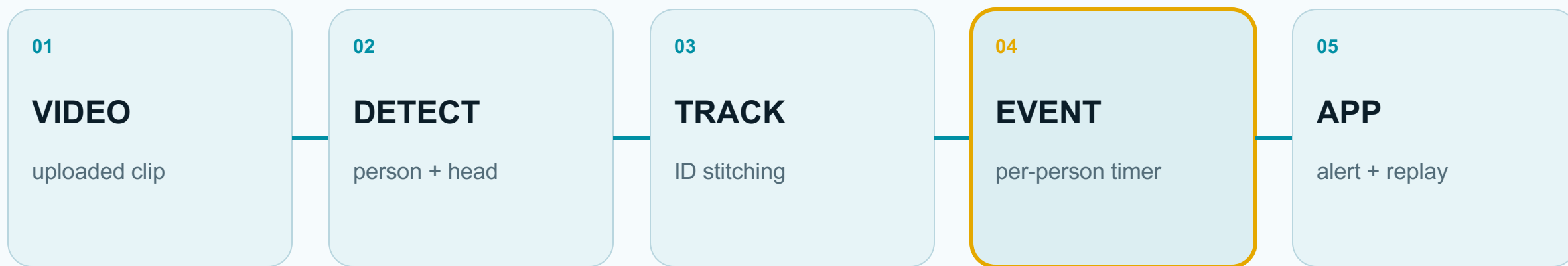
**P1 head below water
for 4.2 seconds**

Alert preference

Yellow + red

Incident replay available

05 Five workstreams, one JSON contract



Contract fields keep every stage testable: frame time, person ID, boxes, head state, confidence, timer state, reason

Fake tracks first • real detector output next • freeze and rehearse last

A timer fails when identity flips

Five MuJoCo scenarios with four simulated people

18

IDs from the stock tracker

4

stable IDs after cleanup,
delayed confirmation, and stitching

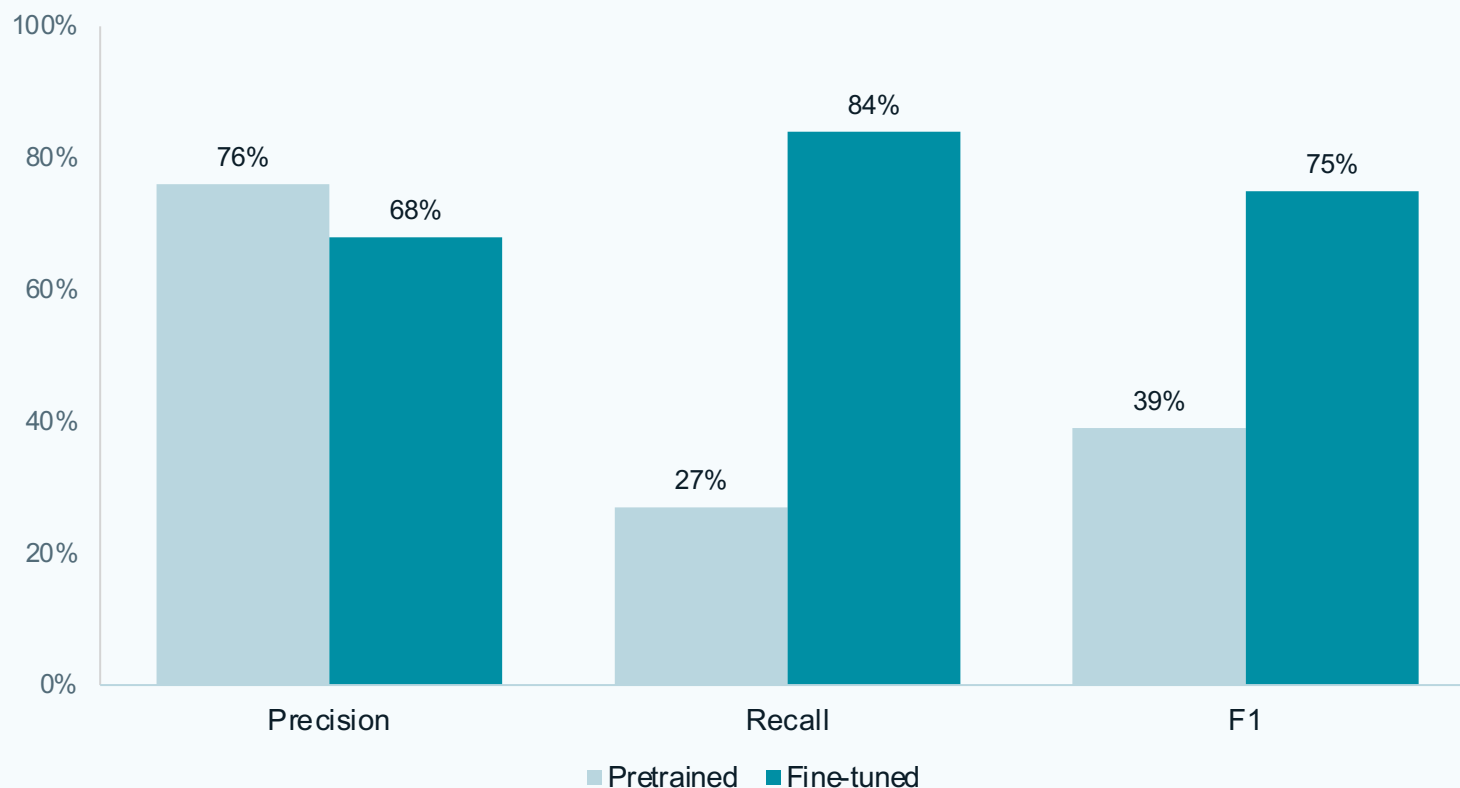
0 switches

The same person keeps the
same timer

Design decision Time head state. Clear water can keep a submerged body visible, so disappearance is only a backup signal.

Fine-tuning found about three times more swimmers

Wave-pool validation: 16 frames, 680 people • IoU 0.5 • confidence 0.25



0.81

AP50 fine-tuned

0.50

AP50-95 fine-tuned

**Best validation mAP
0.512
at epoch 25**

Precision dipped slightly. Recall and F1 rose sharply.

Wave-pool validation only. These results do not claim backyard-pool accuracy.

Simulation tested the pipeline without filming danger

MuJoCo

Exact person boxes and stored head height

Five scripted scenarios gave deterministic labels for tracking and timer tests.

SILENT SINK

COLLAPSE

ENTRY

Isaac Sim

More realistic people, moving water, held-out clip

The second simulator challenged visual realism while preserving safe, repeatable cases.

RESURFACING

MULTI-PERSON OVERLAP

Simulation validates integration. It does not establish real-pool accuracy.

Evaluation errors caught early

RENDER

A batch omitted water

Scores looked artificially high

INPUT

A test script swapped color channels

Detector results became unreliable

LABEL

The first head-under rule was too strict

Ordinary swimmers counted as submerged

Labels were rebuilt from stored head height. Every reported number was rerun.

The claim is deliberately narrow

LIFE VISION DOES

Track one person with one timer

Measure head above or below water

Explain the reason for each alert

Run the current pipeline locally

LIFE VISION DOES NOT

Diagnose drowning

Replace responsible adult supervision

Treat disappearance as the primary signal

Claim real backyard-pool accuracy

Entry detection is separate and only runs when the user arms it.

11 Safe staged clips are next

01

CAPTURE

Record safe staged pool clips

02

LABEL

Head above or below water

03

SPLIT

Train and test by recording session

04

REPORT

Delay, misses, false warnings, ID errors

Field metrics

Detection delay • missed events • false warnings per monitored hour • identity errors

Source path: uploaded video first, then webcam and RTSP. Authorized Ring and Nest integrations can follow. The overlay stays in this app.

